Sahaj Bhadja

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Summary

Software Engineer and USC MSCS graduate with 2+ years shipping AI-powered, full-stack products. Experienced in **end-to-end product ownership**, customer workflows, React, TypeScript, Python, APIs, real-time systems, and reliable production delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Operating Systems, Software Engineering, Database Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Data Structures, Web Technologies, Machine Learning

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*
- Owned multi-tenant product improvements for 500+ users, translating operational needs into normalized workflows and indexed dashboards while reducing latency from 1.2 seconds to 350 milliseconds through **customer-focused iteration**.
- Restructured fault-tolerant AI orchestration with explicit states, background workers, and failure isolation, tripling throughput for 1K+ concurrent jobs while sustaining 99%+ **production reliability** across 5K+ monthly tasks.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*
- Built React and Flask product workflows for uploads, grading history, and feedback, supporting 150+ instructors and 10K+ daily requests through **full-stack ownership**, REST APIs, and rapid user-driven iteration.
- Developed concurrent AI processing with model selection, OCR, provider fallbacks, and failure isolation, handling 1K+ files per batch while **automated validation** reliably reduced instructor feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*
- Designed reusable integration modules from stakeholder requirements, improving API response times 30%, reducing technical debt 20%, and increasing **product maintainability** across distributed Windows, Linux, and macOS deployments.
- Automated 20+ unit and integration tests with PyTest, Mockito, Jenkins, and virtual-machine builds, supporting **independent delivery**, defect diagnosis, customer feedback, and reliable releases across three operating systems.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*
- Built eight MERN services for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ distributed records through **real-time operator workflows** while reducing staged production conflicts 33% during releases.
- Developed responsive React interfaces and WebSocket integrations from user requirements, while Jenkins CI/CD and Docker staging reduced **deployment time** 50% and maintained 95%+ validation across production releases.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*
- Built React and TypeScript booking workflows for scheduling, reservations, pricing, and payment status across 50K+ records, delivering sub-200-millisecond **customer-facing features** through iterative client demonstrations and stakeholder feedback.
- Translated client requirements into modular architecture for 12+ integrations, aligning REST APIs, Firebase synchronization, webhooks, and cross-platform delivery while reducing **design errors** 25% through early product planning.

Skills

Languages: TypeScript, JavaScript, Python, Java, SQL, C++, HTML5, CSS3, Bash
Product Engineering: React, Next.js, Node.js, FastAPI, Flask, REST APIs, WebSockets, Full-Stack Development, Product Discovery, Customer Interviews, Roadmapping, Prioritization, Prototyping, User Analytics
AI & Data: LLM APIs, Conversational AI, Prompt Engineering, Speech Processing, RAG, Vector Search, PostgreSQL, SQL Server, MySQL, MongoDB, Redis, Data Modeling, Real-Time Analytics
Quality & Delivery: Git, Code Review, PyTest, Jest, Unit Testing, Integration Testing, CI/CD, GitHub Actions, Jenkins, Docker, Kubernetes, AWS, Azure, Vercel, Agile/Scrum, Debugging, Observability, Production Support

Projects

CivicCall AI | *TypeScript, React, FastAPI, WebSockets, Twilio, Whisper, PostgreSQL*
- Designed a **public-safety call assistant** that transcribes conversations, classifies non-emergency requests, surfaces configurable response checklists, and transfers urgent cases with structured caller context through real-time dispatcher workflows.

Big-Brain | *Python, LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, REST APIs*
- Built an AI knowledge platform combining **retrieval-augmented generation**, semantic search, vector storage, and REST APIs to deliver grounded responses from domain documents through scalable cloud-hosted application workflows.

HMIS – Hospital Management Information System Live | *MERN, TypeScript, ABDM APIs, Sentry*
- Built consent-driven **critical-service workflows** for patient history, appointments, alerts, and operational monitoring, reducing administrative workload 30% while supporting 100+ daily appointments through accessible interfaces and reliable integrations.